CHEMISTRY
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FULL PAPER

DOI: 10.1002/chem.201202835

Amide-Controlled Highly Selective Catalytic Borylcupration of Allenes

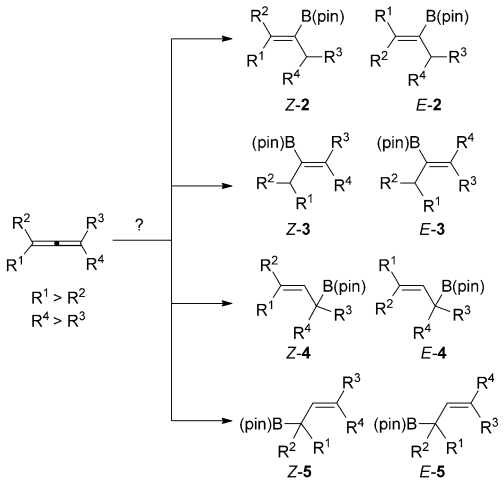
Weiming Yuan,[a] Xue Zhang,*[b] Yihua Yu,[c] and Shengming Ma*[a, b]

Abstract: A novel copper-catalyzed, highly regio- and stereoselective borylcupration of substituted 2,3-allenamides with bis(pinacolato)diboron producing *Z*-β-borylated β,γ-unsaturated enoamides has been demonstrated. Due to the unique effect of the amide-group, perfect regio- and stereoselectivity and good to excellent yields have been achieved, which were rationalized by a DFT study.

Keywords: allenes · amides · boron · copper · regioselectivity · stereoselectivity

Introduction

Organoboranes are a class of highly useful intermediates that are widely used in organic synthesis because of their potential use in the construction of complex molecules.[1–4] In addition to the traditional hydroboration of terminal alkenes and alkynes with boranes,[5] much attention has been paid to the synthesis of organoboranes from the Cu-catalyzed borylcupration of unsaturated hydrocarbons with bis(pinacolato)diboron.[6–11] Yun and Hoveyda and co-workers reported the copper-catalyzed reaction of terminal alkynes or aryl-substituted nonterminal alkynes with $B_2(pin)_2$, affording stereodefined 1-alkenyl boronates.[6] The challenge for such syntheses remains the limited range of substrates that allow control of regioselectivity and inclusion of synthetically attractive functional groups. For the Cu-catalyzed borylcupration reaction of multisubstituted allenes, a formidable challenge is to control the regio- and stereoselectivity because, in principle, it is possible to produce eight regio- and stereoisomers (Scheme 1). In 2011, Santos and co-workers first reported the copper-catalyzed β-borylation of allenoates by using a preactivated diboron reagent (PDIPA = pinacolato diisopropanolaminato diboron) and CF_3CH_2OH as proton source, albeit with a limited substrate scope and low



Scheme 1. Challenges for the hydroboration of multisubstituted allenes.

to moderate yields (Scheme 2a).[12] We developed highly regio- and stereoselective copper-catalyzed borylcuprations of aryl-substituted allenes with bis(pinacolato)diboron to produce 2-alken-2-yl boronates as the only product with exclusive *Z* geometry in the presence of tris(*p*-methoxyphenyl)phosphine as ligand. When the bidentate phosphine, 2,2′-bis(diphenylphosphino)biphenyl (BIPHEP), was used, the regioselectivity switched; alkyl, benzyl, or aryl-substituted allenes all afforded the 1-alken-2-yl boronates highly selectively (Scheme 2b).[13] Hoveyda and co-workers reported the hydroboration of a specific substrate, 4-(*o*-methoxyphenyl)-1,2,5-hexatriene, by using a NHC–Cu complex as catalyst with a regioselectivity of 9:1 (Scheme 1c).[14] It should be noted that moisture- and air-sensitive NaO*t*Bu was used as base in these two reactions.[13,14] Thus, there are some challenges that require further attention. Here, we wish to report our recent observation that by introducing a very common and synthetically attractive directing functional group, i.e., amide, the issues of regio- and stereoselectivity of borylcupration of allenes have been addressed under mild conditions with K_2CO_3 as base, producing useful

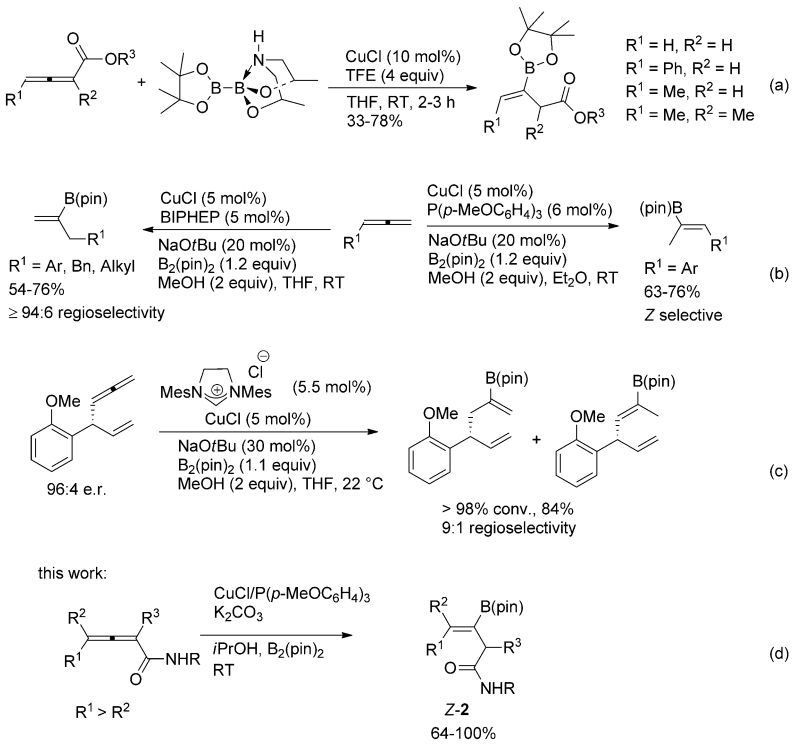
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Supporting information for this article is available on the WWW under http://dx.doi.org/10.1002/chem.201202835.

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Scheme 2. Previous reports and the new approach.

Z-alkenyl boronates containing an amide functional group exclusively by applying $P(p\text{-}MeOC_6H_4)_3$ as the ligand (Scheme 2d).

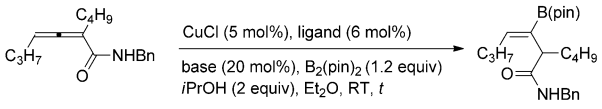
Results and Discussion

After numerous trials with allenyl ketones, carboxylic acids, and esters, we started to observe some interesting results with 2,3-allenamide **1a**. Firstly, we observed that although the CuCl-catalyzed reaction with triphenyl phosphine as the ligand did not proceed (Table 1, entry 1), when 20 mol% K_3PO_4 was added, to our delight, the reaction afforded **2a** in 85% yield with *Z*-geometry exclusively at room temperature (Table 1, entry 2); the reaction with K_2CO_3 afforded similar results (Table 1, entry 3). The use of other bases did not lead to any further improvement in the results (Table 1, entries 4 and 5). Furthermore, it was exciting to observe that by increasing the electron-donating ability of the applied ligand, the reaction proceeded much faster and afforded the product *Z*-**2a** in higher yields (Table 1, entries 6–8); the use of K_2CO_3 (5 mol%) was sufficient for such a transformation, although a longer reaction time (8 h) was required (Table 1, entry 9). Thus, we chose 5 mol% CuCl, 6 mol% $P(p\text{-}MeOC_6H_4)_3$, 20 mol% K_2CO_3, and 2 equivalents of *i*PrOH in Et_2O at room temperature as the standard conditions for the borylcupration reaction.

With the optimized conditions in hand, we observed that different 2,4-dialkyl-substituted allenamides afforded the tri-substituted vinylboronates *Z*-**2** exclusively in good to excellent yields (Table 2, entries 1–8); with R^1 being an *iso*-alkyl group (entries 4–7) or a longer carbon chain (entry 8), the yields were somewhat lower. The reaction of 4-benzyl-substituted allenamide **1h** also produced *Z*-**2h** in good yield (entry 9); R^1 or R^2 may also be an aryl- or heteroaryl group (entries 10–13). For 2-unsubstituted 2,3-pentadienamide **1m**, the *Z*/*E* selectivity was lower. After screening, it was observed that the reaction in THF afforded the product *Z*-**2m** in 90% yield with a high regio- and stereoselectivity (entry 15). The geometry of the remaining C=C bond in **2d** and **2g** was determined to be *Z*, based on NOESY experiments. In addition, the reaction of **1d** proceeded smoothly to afford *Z*-**2d** on a one gram scale in 86% yield (entry 5).

The borylcupration of 4,4-disubstituted allenamides also proceeded smoothly in almost quantitative yields irrespective of whether the N-protecting group was Bn, Ph, or *n*Bu (Scheme 3). Even for 4-methyl-4-*tert*-butyl- or 4-methyl-4-phenyl-substituted allenamides **1q** and **1r**, the reaction afforded the corresponding products *Z*-**2q** and *Z*-**2r** as the only isomer in good yields. The structure of *Z*-**2q** was fur-

Table 1. Optimization of reaction conditions for copper-catalyzed regio- and stereoselective borylcupration of 2,3-allenamides.



1a *Z*-**2a**

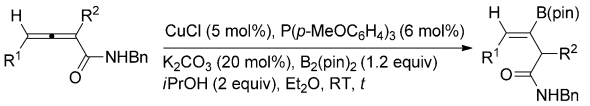
Entry	Base	Ligand	*t* [h]	Yield *Z*-**2a** [%][a]
1	–	PPh_3	11	0
2	K_3PO_4	PPh_3	2.5	85
3	K_2CO_3	PPh_3	2.5	83
4	NaOAc	PPh_3	7	84 (4)[b]
5	Na_2CO_3	PPh_3	11	75 (20)[b]
6	K_2CO_3	$P(p\text{-}MeOC_6H_4)_3$	2.5	91
7	K_2CO_3	TFP	2	90
8	K_2CO_3	$P(o\text{-tol})_3$	2	91
9	K_2CO_3[c]	$P(p\text{-}MeOC_6H_4)_3$	8	89

[a] The starting material **1a** was converted completely, and the yield of *Z*-**2a** was determined by ^1H NMR spectroscopic analysis using 1,3,5-trimethylbenzene as internal standard. [b] The amount of recovered starting material is given in parentheses. [c] K_2CO_3 (5 mol%) was added. TFP = tri(2-furyl)phosphine. $P(o\text{-tol})_3$ = tri(*o*-tolyl)phosphine.

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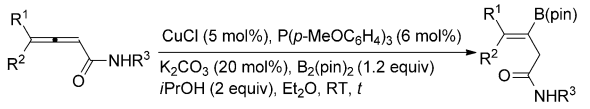
Table 2. CuCl-catalyzed addition of bis(pinacolato)diboron to 2,4-disubstituted 2,3-allenamides.



1 Z-2

Entry[a]	R^1	R^2	1	t [h]	Yield [%][b]	2
1	Me	*n*Bu	**1a**	2	94	*Z*-**2a**
2	Et	*n*Bu	**1b**	2	92	*Z*-**2b**
3	*n*-Pr	*n*Bu	**1c**	2	90	*Z*-**2c**
4	*i*Pr	*n*Bu	**1d**	3	87	*Z*-**2d**
5[c]	*i*Pr	*n*Bu	**1d**	2.5	86	*Z*-**2d**
6	*i*Bu	*n*Bu	**1e**	3	86	*Z*-**2e**
7	cyclohexyl	*n*Bu	**1f**	2.5	84	*Z*-**2f**
8	nC_5H_{11}	nC_8H_{17}	**1g**	2.5	86	*Z*-**2g**
9	Bn	*n*Bu	**1h**	2	87	*Z*-**2h**
10	Ph	Me	**1i**	2	77	*Z*-**2i**
11	*p*-$MeOC_6H_4$	nC_8H_{17}	**1j**	2	82	*Z*-**2j**
12	2-thenyl	*n*Bu	**1k**	2	64	*Z*-**2k**
13	Me	*p*-MeC_6H_4	**1l**	2	67	*Z*-**2l**
14	Me	H	**1m**	2	54[d]	*Z*-**2m**/*E*-**2m**
15[e]	Me	H	**1m**	2.5	90	*Z*-**2m**

[a] Reaction conditions: 2,3-allenamide **1** (0.5 mmol), bis(pinacolato)diboron (0.6 mmol), CuCl (5 mol%), P(*p*-$MeOC_6H_4$)$_3$ (6 mol%), K_2CO_3 (20 mol%), *i*PrOH (1.0 mmol), Et_2O (2 mL), RT. [b] Yield upon isolation. [c] The reaction was carried out on a one gram scale. [d] The yield was determined by ^{1}H NMR spectroscopic analysis using 1,3,5-trimethylbenzene as internal standard and *Z*-**2m**/*E*-**2m** = 83:17. [e] THF was used as solvent.



1n: R^1 = Me, R^2 = Me, R^3 = Bn
1o: R^1 = Me, R^2 = Me, R^3 = Ph
1p: R^1 = Me, R^2 = Me, R^3 = *n*Bu
1q: R^1 = Me, R^2 = *t*Bu, R^3 = Bn
1r: R^1 = Me, R^2 = Ph, R^3 = Bn

2n: 2.5 h, 97%
2o: 2 h, 100%
2p: 2 h, 98%
Z-**2q**: 3 h, 70%
Z-**2r**: 3.5 h, 76%

Scheme 3. Borylcupration of 4,4-disubstituted allenamides.

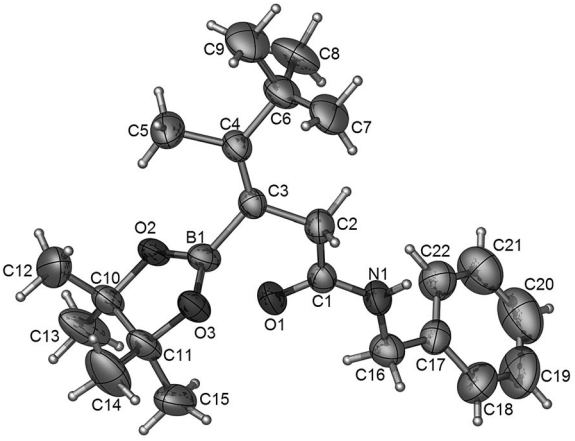
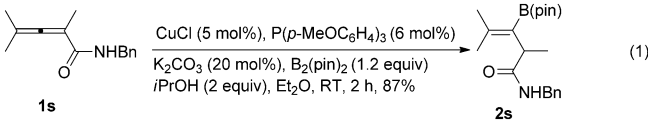


Figure 1. ORTEP representation of *Z*-**2q**.

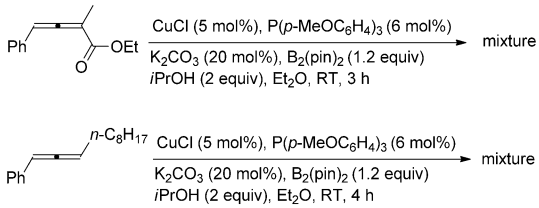
ther confirmed by X-ray single-crystal diffraction studies (Figure 1).[15]

Finally, the reaction could also be extended to 2,4,4-trisubstituted allenamide **1s**, affording **2s** in 87% yield [Eq. (1)].



1s 2s (1)

In contrast, 2,4-disubstituted 2,3-allenoate or 1,3-disubstituted phenyl-1,2-decadiene afforded a mixture with poor regioselectivity[16] (Scheme 4), indicating the importance of the amide group.



Scheme 4. Borylcupration of 2,4-disubstituted 2,3-allenoate or 1,3-disubstituted phenyl-1,2-decadiene.

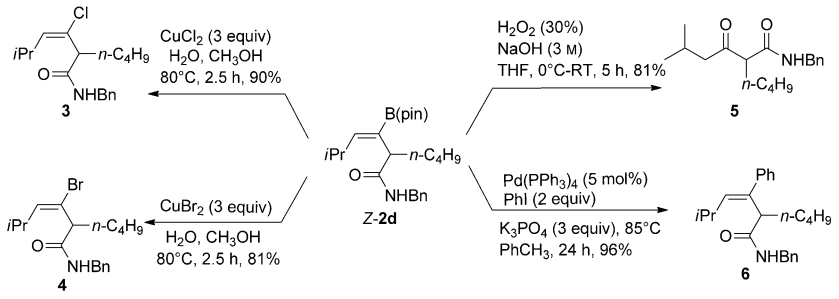
To demonstrate the utility of the highly regio- and stereoselective borylcupration reaction, halogenation of *Z*-**2d** was attempted by using $CuCl_2$ or $CuBr_2$ to produce the corresponding vinyl chloride and vinyl bromide, respectively, in excellent yields (Scheme 5).[17] Subsequently, we also attempted the hydroxylation of *Z*-**2d**, affording β-ketoamide **5** in 81% yield.[9c] In addition, the Suzuki coupling of *Z*-**2d** with iodobenzene produced the coupled product in 96% yield without loss of geometrical purity.[18]

To unveil the mechanism, the reaction of **1n** was then conducted in the presence of *i*PrOD. It was interesting to observe that the reaction afforded 96% yield of [D]-**2n** with 50% deuterium incorporation at the allylic position and 46% at the nitrogen atom; in fact, it was observed that the reaction of **1t** without NH afforded [D]-**2t** with 84% deuterium incorporation at the allylic position (Scheme 6).

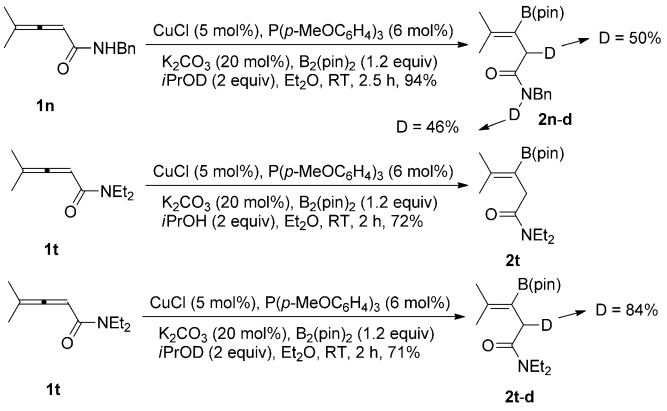
It is generally believed that phosphine copper(I) boryl complexes [(PR_3)Cu(boryl)] are the active species in catalytic borylcupration reactions.[6] The active species can be easily generated from the phosphine ligand coordination plus a transmetalation reaction between CuCl and $B_2(pin)_2$. According to the experimentally observed regioselectivity, the boryl moiety installs exclusively on the central allene carbon. Therefore, the possibilities of two alternative pathways, related to this β-borylation regioselectivity, were investigated by DFT calculations[19] based on the substrate model *N*-benzylbuta-2,3-dienamide **1** and the model phosphine copper(I) boryl complex [(PMe_3)Cu(BR)] (R = $-OCH_2CH_2O-$), in which the aryl groups in the phosphine

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Scheme 5. Versatile transformations of the borylated product.



Scheme 6. Control experiments in the presence of *i*PrOH or *i*PrOD.

ligand and the methyl groups in the Bpin were replaced by CH_3 and H, respectively. As shown in Figure 2,[20] coordination of the active species [(PMe_3)-Cu(BR)] to the two C=C bonds of the model allenamide **1** were considered separately. The metal-η^2-allene intermediate **1A** was found to be slightly lower in free energy and electronic energy than **1B**. From **1A** and **1B**, the coordinated C=C bond undergoes an insertion via transition states $\mathbf{TS1_A}$ and $\mathbf{TS1_B}$ to give the β-boryl alkenyl complexes **2A** and **2B**, respectively. Intermediate **2B** was more stable than **2A** by 4.5 kcal mol^{-1}, probably due to coordination of the C=O moiety with the copper(I) metal center in **2B** (Cu–O bond length 2.480 Å). Nevertheless, transition state $\mathbf{TS1_B}$, which leads to the formation of the more stable intermediate **2B**, was calculated to hold a higher barrier for insertion than $\mathbf{TS1_A}$. In $\mathbf{TS1_A}$, the C=O moiety coordinates with the copper(I) metal center with a Cu–O bond length of 2.697 Å, resulting in a lower barrier of $\mathbf{TS1_A}$ than that of $\mathbf{TS1_B}$ by 4.6 kcal mol^{-1} (8.2 vs. 12.8 kcal mol^{-1}). The kinetically favorable intermediate **2A** quickly isomerizes to the more stable intermediate **2B**, with a very small barrier of 1.9 kcal mol^{-1}, through metal migration from the α-position to the γ-position. Finally, γ-protonolysis of the allylic C–Cu bond in **2B** or **2A** by MeOH produces

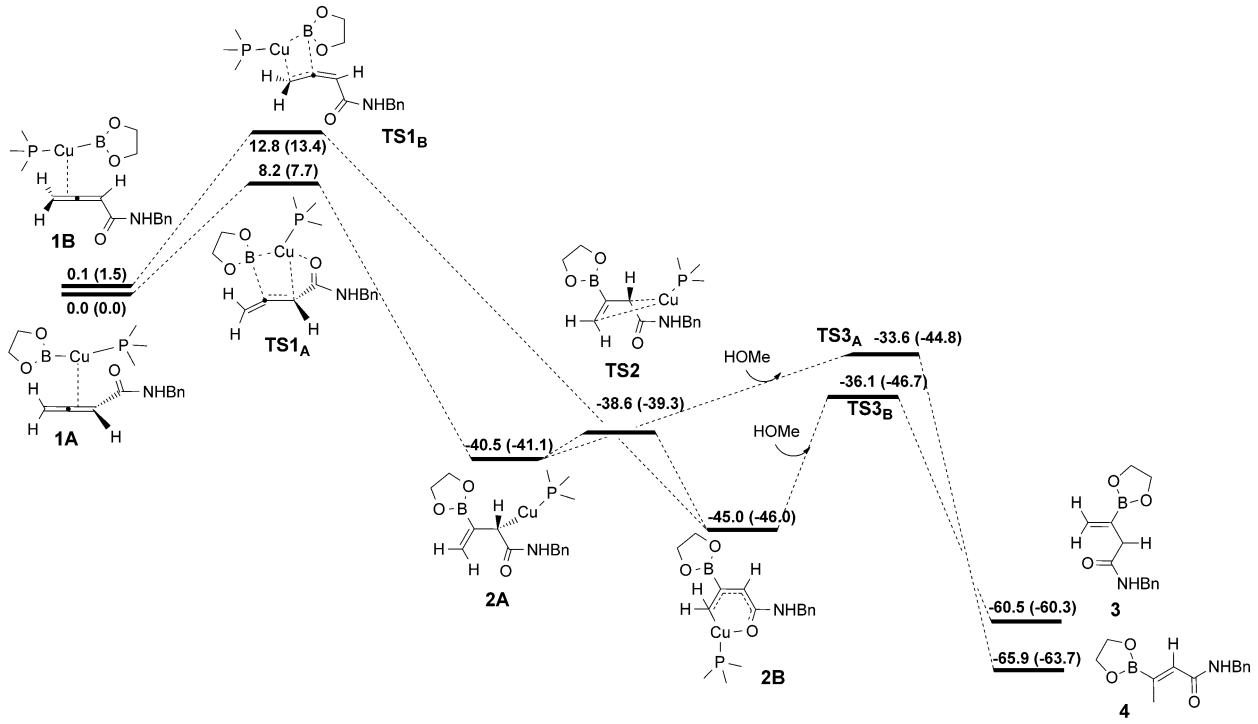


Figure 2. Energy profiles calculated for possible pathways of the borylation of allenamide **1** catalyzed by [(PMe_3)Cu(Bpin)]. The relative free energies and electronic energies (in parentheses) are given in kcal mol^{-1}.

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the major product **3** or its regioisomer **4**, and generates copper-alkoxide, which is then followed by a metathesis step with $B_2(pin)_2$ to regenerate the active species [$(PMe_3)Cu(BR)$] and complete the catalytic cycle. Thus, it is the stability of **2B** over **2A** that controls the regioselectivity of this borylcupration reaction. It is clear that the reaction predominantly takes place with the amide-substituted C=C bond and the rate-limiting step is insertion of the coordinated allenamide into the Cu–B bond (Figure 2).

The stereoselectivity of the reaction was further rationalized by DFT calculations on the model substrate *N*-benzylpenta-2,3-dienamide (**1m**). Only the two competing transition states of the rate-limiting insertion step were further considered, which are denoted as $\mathbf{TS1_A_}\boldsymbol{Z}$ and $\mathbf{TS1_A_}\boldsymbol{E}$ (Figure 3), respectively.[21] As shown in Figure 3, $\mathbf{TS1_A_}\boldsymbol{Z}$, which leads to the formation of the *Z* isomer, was found to be more stable in free energy than $\mathbf{TS1_A_}\boldsymbol{E}$ by 3.8 kcal mol^{-1}, thus, accounting for the observed overwhelming formation of *Z* products. In $\mathbf{TS1_A_}\boldsymbol{E}$, the boryl group suffers a severe steric interaction with the γ-substituent of the allenamide (H···H distance of only 2.705 Å). The angle α must open up to reduce the steric interaction, resulting in reduced stability of $\mathbf{TS1_A_}\boldsymbol{E}$. Therefore, boryl addition always occurs on the opposite side of the γ-substituent to produce the more stable *Z* isomer.

Based on these observations and on the results of the DFT study, a catalytic cycle is proposed as shown in Scheme 7. Firstly, the copper-boryl complex PCu-B(pin) adds to the C=C double bond next to the amide group, with boron always connected to the middle carbon atom. Subsequently, Cu migrates from the α-position to the γ-position in the allylic unit, forming the more stable six-membered cyclic intermediate with chelation between Cu and C=O. Finally, γ-protonolysis of the allylic C–Cu bond with *i*PrOH produces the final product *Z*-**2** due to the severe steric interaction between the boryl group and the larger γ-substituent of the allenamide.

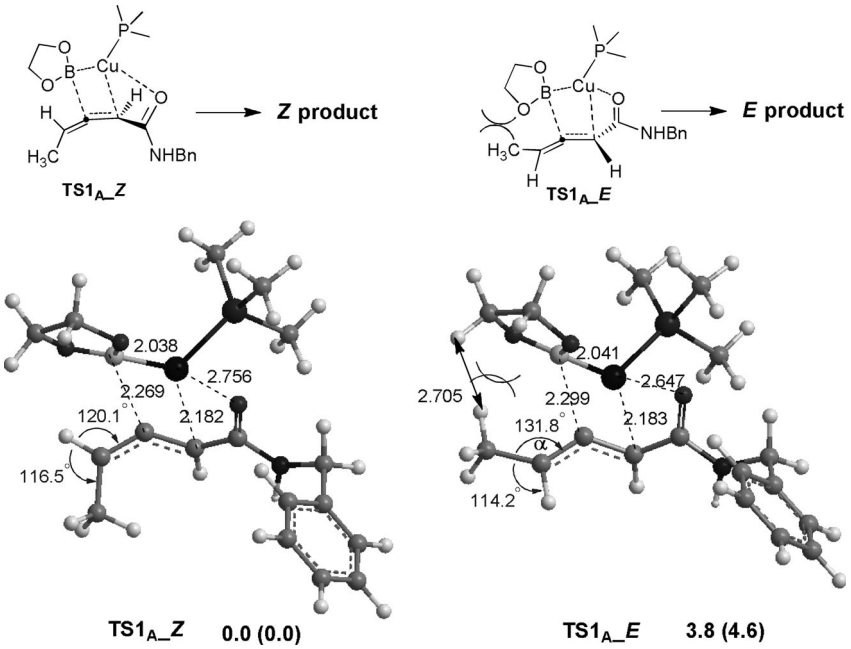
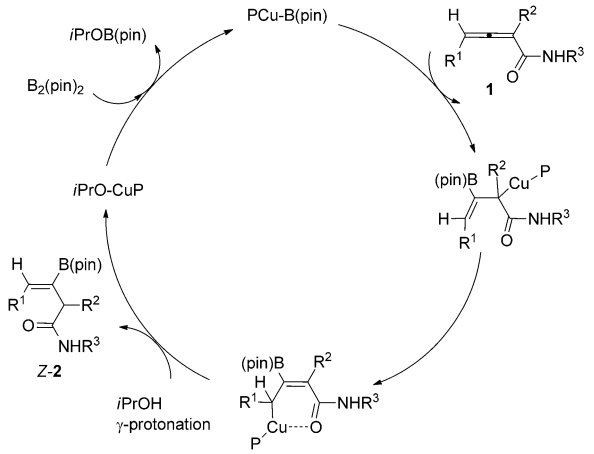


Figure 3. The calculated transition states with selected structural parameters (distances in Å) for the insertion of the model allenamide **1m** into the Cu–B bond. The relative free energies and electronic energies (in parentheses) are given in kcal mol^{-1}.



Scheme 7. Summary of the catalytic cycle for the borylation of allenamide.

Conclusion

We have developed an efficient copper-catalyzed borylcupration of 2,3-allenamides by using a catalytic amount of CuCl, P(*p*-$MeOC_6H_4)_3$, and K_2CO_3 as well as *i*PrOH as the proton source to produce the β-borylated β,γ-unsaturated enoamides with exclusive *Z* C=C bond geometry. The excellent regio- and stereoselectivity has been validated by DFT studies, which may be used as the principle for tuning the selectivities in such borylcuprations of other functionalized allenes. These otherwise difficult to prepare borylated products are highly functionalized and thus, are very useful intermediates for subsequent elaboration of the C–B, C=C, and amide functionalities for the synthesis of complex structures. Further studies examining the synthetic application of this method and the mechanism are underway in this laboratory.

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Experimental Section

General methods: All reactions were carried out in oven-dried Schlenk tubes under an argon atmosphere. Et_2O and THF were distilled over sodium wire using benzophenone as the indicator under argon atmosphere. *i*PrOH was distilled using magnesium turnings as drying agent under argon.

Typical reaction procedure for the copper-catalyzed borylcupration of 2,3-allenamides—*Preparation of N-benzyl-2-butyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pent-3-(Z)-enamide (Z-**2a**)*: To an oven-dried Schlenk tube equipped with a magnetic stirring bar were added sequentially bis(pinacolato)diboron (153.0 mg, 0.60 mmol), P(*p*-$MeOC_6H_4)_3$ (10.6 mg, 0.03 mmol), CuCl (2.5 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1a** (123.0 mg, 0.50 mmol), *i*PrOH (78 μL, $d=0.784\ g\,mL^{-1}$, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) under argon. The mixture was stirred at RT for 2 h and monitored by TLC. Upon completion, the resulting mixture was filtered through a short column of silica gel eluted with Et_2O and concentrated. The residue was purified by chromatography on silica gel (petroleum ether/ethyl acetate = 10:1) to afford *Z*-**2a** (176.6 mg, 94%) as a liquid. ^1H NMR (300 MHz, $CDCl_3$): $\delta=7.35$–7.19 (m, 5H; Ar-H), 6.77 (brs, 1H; NH), 6.62 (q, $J=6.8$ Hz, 1H; CH=C), 4.47–4.33 (m, 2H; CH_2Ph), 3.44 (dd, $J_1=9.8$ Hz, $J_2=5.9$ Hz, 1H; COCH), 2.05–1.90 (m, 1H; CH*H*), 1.86–1.68 (m, 4H; $=CH_3$ and C*H*H), 1.38–1.08 (m, 16H; $2\times CH_2$ and B(pin)), 0.87 ppm (t, $J=7.1$ Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=174.4$, 144.8, 138.7, 128.4, 127.7, 127.1, 83.5, 47.7, 43.5, 31.1, 29.9, 24.8, 24.1, 22.5, 14.7, 13.9 ppm; IR (neat): $\tilde{\nu}=3369$, 2977, 2929, 1656, 1535, 1455, 1409, 1384, 1344, 1305, 1215, 1139 cm^{-1}; MS (70 eV, EI): *m/z* (%): 371 (14.63) $[M]^+$, 91 (100); HRMS: *m/z* calcd for $C_{22}H_{34}BNO_3$: 371.2632 $[M]^+$; found: 371.2631.

*Preparation of (Z)-N-benzyl-2-butyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)hex-3-(Z)-enamide (Z-**2b**)*: The reaction of bis(pinacolato)diboron (152.0 mg, 0.60 mmol), P(*p*-$MeOC_6H_4)_3$ (10.4 mg, 0.03 mmol), CuCl (2.4 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1b** (128.8 mg, 0.50 mmol), *i*PrOH (78 μL, $d=0.784\ g\,mL^{-1}$, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2b** (177.4 mg, 92%) after purification (petroleum ether/ethyl acetate = 10:1) as a liquid. ^1H NMR (300 MHz, $CDCl_3$): $\delta=7.35$–7.20 (m, 5H; Ar-H), 6.81 (brs, 1H; NH), 6.50 (t, $J=7.1$ Hz, 1H; CH=C), 4.40 (d, $J=5.4$ Hz, 2H; CH_2Ph), 3.41 (dd, $J_1=9.9$ Hz, $J_2=5.7$ Hz, 1H; COCH), 2.34–2.18 (m, 2H; CH_2), 2.04–1.85 (m, 1H; C*H*H), 1.83–1.66 (m, 1H; CH*H*), 1.38–1.07 (m, 16H; $2\times CH_2$ and B(pin)), 1.01 (t, $J=7.5$ Hz, 3H; CH_3), 0.87 ppm (t, $J=7.2$ Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=174.6$, 152.1, 138.7, 128.4, 127.7, 127.1, 83.5, 48.1, 43.5, 31.1, 30.0, 24.8, 24.1, 22.5, 22.1, 13.9, 13.5 ppm; IR (neat): $\tilde{\nu}=3370$, 2961, 2930, 2872, 1656, 1536, 1456, 1405, 1372, 1348, 1303, 1216, 1140, 1028 cm^{-1}; MS (70 eV, EI): *m/z* (%): 385 (4.97) $[M]^+$, 91 (100); HRMS: *m/z* calcd for $C_{23}H_{36}BNO_3$: 385.2788 $[M]^+$; found: 385.2790.

*Preparation of (Z)-N-benzyl-2-butyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)hept-3-(Z)-enamide (Z-**2c**)*: The reaction of bis(pinacolato)diboron (152.4 mg, 0.60 mmol), P(*p*-$MeOC_6H_4)_3$ (10.6 mg, 0.03 mmol), CuCl (2.5 mg, 0.025 mmol), K_2CO_3 (13.8 mg, 0.10 mmol), **1c** (136.0 mg, 0.50 mmol), *i*PrOH (78 μL, $d=0.784\ g\,mL^{-1}$, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2c** (180.1 mg, 90%) after purification (petroleum ether/ethyl acetate = 10:1) as a liquid. ^1H NMR (300 MHz, $CDCl_3$): $\delta=7.35$–7.19 (m, 5H; Ar-H), 6.81 (brs, 1H; NH), 6.50 (t, $J=7.2$ Hz, 1H; CH=C), 4.39 (d, $J=5.7$ Hz, 2H; CH_2Ph), 3.42 (dd, $J_1=9.8$ Hz, $J_2=5.9$ Hz, 1H; COCH), 2.28–2.16 (m, 2H; CH_2), 2.03–1.87 (m, 1H; CH*H*), 1.83–1.68 (m, 1H; C*H*H), 1.52–1.06 (m, 18H; $3\times CH_2$ and B(pin)), 0.96–0.82 ppm (m, 6H; $2\times CH_3$); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=174.5$, 150.4, 138.7, 128.4, 127.6, 127.1, 83.5, 48.2, 43.4, 31.2, 30.9, 30.0, 24.7, 24.0, 22.5, 22.1, 13.91, 13.88 ppm; IR (neat): $\tilde{\nu}=3370$, 2958, 2929, 2871, 1656, 1535, 1455, 1405, 1373, 1349, 1304, 1216, 1140 cm^{-1}; MS (70 eV, EI): *m/z* (%): 399 (13.40) $[M]^+$, 91 (100); HRMS: *m/z* calcd for $C_{24}H_{38}BNO_3$: 399.2945 $[M]^+$; found: 399.2946.

*Preparation of (Z)-N-benzyl-2-butyl-5-methyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)hex-3-(Z)-enamide (Z-**2d**)*: The reaction of bis(pinacolato)diboron (153.0 mg, 0.60 mmol), P(*p*-$MeOC_6H_4)_3$ (10.6 mg, 0.03 mmol), CuCl (2.5 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1d** (135.7 mg, 0.50 mmol), *i*PrOH (78 μL, $d=0.784\ g\,mL^{-1}$, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2d** (174.1 mg, 87%) after purification (petroleum ether/ethyl acetate = 10:1) as a liquid. ^1H NMR (300 MHz, $CDCl_3$): $\delta=7.35$–7.18 (m, 5H; Ar-H), 6.85 (brs, 1H; NH), 6.30 (d, $J=9.6$ Hz, 1H; CH=C), 4.48–4.33 (m, 2H; CH_2Ph), 3.42 (dd, $J_1=10.2$ Hz, $J_2=6.0$ Hz, 1H; COCH), 2.94–2.76 (m, 1H; CH), 2.02–1.88 (m, 1H; CH*H*), 1.83–1.67 (m, 1H; C*H*H), 1.39–1.06 (m, 16H; $2\times CH_2$ and B(pin)), 0.98 (d, $J=6.6$ Hz, 3H; CH_3), 0.97 (d, $J=6.3$ Hz, 3H; CH_3), 0.87 ppm (t, $J=7.1$ Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=174.6$, 157.2, 138.7, 128.4, 127.6, 127.1, 83.5, 48.3, 43.4, 31.2, 30.0, 27.6, 24.7, 24.1, 22.50, 22.47, 22.3, 13.9 ppm; IR (neat): $\tilde{\nu}=3374$, 2960, 2869, 1657, 1539, 1466, 1371, 1348, 1303, 1271, 1230, 1142 cm^{-1}; MS (70 eV, EI): *m/z* (%): 399 (8.52) $[M]^+$, 91 (100); HRMS: *m/z* calcd for $C_{24}H_{38}BNO_3$: 399.2945 $[M]^+$; found: 399.2942.

*Preparation of (Z)-N-benzyl-2-butyl-5-methyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)hex-3-(Z)-enamide (Z-**2d**) on a one gram scale*: The reaction of bis(pinacolato)diboron (1.1201 g, 4.42 mmol), P(*p*-$MeOC_6H_4)_3$ (77.6 mg, 0.22 mmol), CuCl (17.9 mg, 0.18 mmol), K_2CO_3 (101.7 mg, 0.74 mmol), **1d** (1.0060 g, 3.68 mmol), *i*PrOH (574 μL, $d=0.784\ g\,mL^{-1}$, 442.3 mg, 7.36 mmol), and Et_2O (10 mL) afforded *Z*-**2d** (1.2731 g, 86%) after purification (petroleum ether/ethyl acetate = 10:1) as a liquid.

*Preparation of (Z)-N-benzyl-2-butyl-6-methyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)hept-3-(Z)-enamide (Z-**2e**)*: The reaction of bis(pinacolato)diboron (152.0 mg, 0.60 mmol), P(*p*-$MeOC_6H_4)_3$ (10.7 mg, 0.03 mmol), CuCl (2.4 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1e** (142.1 mg, 0.50 mmol), *i*PrOH (78 μL, $d=0.784\ g\,mL^{-1}$, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2e** (176.2 mg, 86%) after purification (petroleum ether/ethyl acetate = 15:1) as a liquid. ^1H NMR (300 MHz, $CDCl_3$): $\delta=7.35$–7.18 (m, 5H; Ar-H), 6.84 (brs, 1H; NH), 6.51 (t, $J=7.2$ Hz, 1H; CH=C), 4.46–4.31 (m, 2H; CH_2Ph), 3.52–3.38 (m, 1H; COCH), 2.20–2.08 (m, 2H; CH_2), 2.05–1.88 (m, 1H; CH*H*), 1.83–1.63 (m, 2H; C*H*H and CH), 1.40–1.02 (m, 16H; $2\times CH_2$ and B(pin)), 0.98–0.78 ppm (m, 9H; $3\times CH_3$); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=174.4$, 149.6, 138.6, 128.3, 127.6, 127.0, 83.4, 48.2, 43.4, 37.8, 31.2, 30.0, 28.3, 24.7, 24.0, 22.50, 22.48, 22.42, 13.9 ppm; IR (neat): $\tilde{\nu}=3372$, 2956, 2871, 1662, 1533, 1466, 1406, 1372, 1351, 1304, 1215, 1141, 1028 cm^{-1}; MS (70 eV, EI): *m/z* (%): 413 (25.42) $[M]^+$, 91 (100); HRMS: *m/z* calcd for $C_{25}H_{40}BNO_3$: 413.3101 $[M]^+$; found: 413.3102.

*Preparation of (Z)-N-benzyl-2-butyl-4-cyclohexanyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)butyl-3-(Z)-enamide (Z-**2f**)*: The reaction of bis(pinacolato)diboron (153.0 mg, 0.60 mmol), P(*p*-$MeOC_6H_4)_3$ (10.6 mg, 0.03 mmol), CuCl (2.5 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1f** (156.0 mg, 0.50 mmol), *i*PrOH (78 μL, $d=0.784\ g\,mL^{-1}$, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2f** (184.0 mg, 84%) after purification (petroleum ether/ethyl acetate = 15:1) as a liquid. ^1H NMR (300 MHz, $CDCl_3$): $\delta=7.34$–7.18 (m, 5H; Ar-H), 6.85 (t, $J=5.0$ Hz, 1H; NH), 6.30 (d, $J=9.3$ Hz, 1H; CH=C), 4.40 (d, $J=5.4$ Hz, 2H; CH_2Ph), 3.42 (dd, $J_1=10.1$ Hz, $J_2=5.9$ Hz, 1H; COCH), 2.58–2.43 (m, 1H; CH), 2.02–1.88 (m, 1H; CH*H*), 1.82–1.53 (m, 6H; $3\times CH_2$), 1.41–1.02 (m, 21H; $4\times CH_2$ and CH and B(pin)), 0.87 ppm (t, $J=7.1$ Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=174.7$, 155.8, 138.7, 128.3, 127.6, 127.0, 83.4, 48.3, 43.4, 37.4, 32.4, 32.3, 31.2, 30.0, 25.8, 25.52, 25.49, 24.7, 24.0, 22.5, 13.9 ppm; IR (neat): $\tilde{\nu}=3370$, 2924, 2852, 1655, 1536, 1452, 1404, 1372, 1344, 1304, 1271, 1228, 1141 cm^{-1}; MS (70 eV, EI): *m/z* (%): 439 (13.88) $[M]^+$, 91 (100); HRMS: *m/z* calcd for $C_{27}H_{42}BNO_3$: 439.3258 $[M]^+$; found: 439.3260.

*Preparation of (Z)-N-benzyl-2-octanyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)non-3-(Z)-enamide (Z-**2g**)*: The reaction of bis(pinacolato)diboron (153.0 mg, 0.60 mmol), P(*p*-$MeOC_6H_4)_3$ (10.7 mg, 0.03 mmol), CuCl (2.4 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1g** (177.5 mg, 0.50 mmol), *i*PrOH (78 μL, $d=0.784\ g\,mL^{-1}$, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2g** (207.6 mg, 86%) after purification (petroleum ether/ethyl acetate = 10:1) as a liquid. ^1H NMR (300 MHz, $CDCl_3$): $\delta=7.34$–7.18 (m, 5H; Ar-H), 6.81 (t, $J=5.3$ Hz, 1H; NH), 6.51 (t, $J=7.2$ Hz, 1H; CH=C), 4.39 (d, $J=5.1$ Hz, 2H; CH_2Ph), 3.42 (dd, $J_1=9.9$ Hz, $J_2=6.0$ Hz, 1H; COCH), 2.38–2.13 (m, 2H; CH_2), 2.05–1.85 (m, 1H; CH*H*), 1.83–1.65 (m, 1H; C*H*H), 1.48–1.18 (m, 18H; $9\times CH_2$), 1.16 (s, 6H; B(pin)), 1.11 (s, 6H; B(pin)), 0.87 ppm (t, $J=6.8$ Hz, 6H; $2\times CH_3$); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=174.4$, 150.6, 138.6, 128.3, 127.6,

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127.0, 83.4, 48.0, 43.4, 31.7, 31.5, 31.4, 29.34, 29.32, 29.1, 28.8, 28.6, 27.7, 24.6, 24.0, 22.5, 22.4, 14.0, 13.8 ppm; IR (neat): $\tilde{\nu}$ = 3371, 2956, 2925, 2856, 1658, 1529, 1456, 1405, 1374, 1349, 1305, 1215, 1140, 1109, 1030 cm^{-1}; MS (70 eV, EI): *m*/*z* (%): 483 (17.45) $[M]^+$, 91 (100); HRMS: *m*/*z* calcd for $C_{30}H_{50}BNO_3$: 483.3884 $[M]^+$; found: 483.3885.

*Preparation of (Z)-N-benzyl-2-butyl-4-benzyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)butyl-3-(Z)-enamide (Z-**2h**)*: The reaction of bis(pinacolato)diboron (151.0 mg, 0.60 mmol), P(*p*-MeOC$_6$H$_4$)$_3$ (10.5 mg, 0.03 mmol), CuCl (2.4 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1h** (158.0 mg, 0.50 mmol), *i*PrOH (78 μL, *d* = 0.784 g mL^{-1}, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2h** (190.3 mg, 87%) after purification (petroleum ether/ethyl acetate = 10:1) as a white solid; m.p. 75–76 °C (*n*-hexane/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.34–7.13 (m, 10H; Ar-H), 6.80 (t, *J* = 5.0 Hz, 1H; NH), 6.63 (t, *J* = 7.2 Hz, 1H; CH=C), 4.48–4.34 (m, 2H; CH_2Ph), 3.63–3.38 (m, 3H; COCH and =CCH_2), 2.08–1.93 (m, 1H; CH*H*), 1.88–1.73 (m, 1H; C*H*H), 1.40–1.22 (m, 4H; 2×CH_2), 1.14 (s, 6H; B(pin)), 1.09 (s, 6H; B(pin)), 0.88 ppm (t, *J* = 7.1 Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 174.2, 148.1, 139.7, 138.6, 128.7, 128.43, 128.42, 127.6, 127.1, 126.0, 83.6, 48.3, 43.5, 35.2, 31.2, 30.0, 24.7, 24.1, 22.5, 13.9 ppm; IR (neat): $\tilde{\nu}$ = 3270, 2974, 2930, 1644, 1559, 1454, 1372, 1299, 1222, 1142, 1028 cm^{-1}; MS (70 eV, EI): *m*/*z* (%): 447 (15.39) $[M]^+$, 91 (100); elemental analysis calcd (%) for $C_{28}H_{38}BNO_3$: C 75.16, H 8.56, N 3.13; found: C 75.29, H 8.31, N 3.01.

*Preparation of (Z)-N-benzyl-2-methyl-4-phenyl-3-(4,4,5,5-tetra-methyl-1,3,2-dioxaborolan-2-yl)butyl-3-(Z)-enamide (Z-**2i**)*: The reaction of bis-(pinacolato)diboron (153.2 mg, 0.60 mmol), P(*p*-MeOC$_6$H$_4$)$_3$ (10.6 mg, 0.03 mmol), CuCl (2.5 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1i** (132.0 mg, 0.50 mmol), *i*PrOH (78 μL, *d* = 0.784 g mL^{-1}, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2i** (150.4 mg, 77%) after purification (petroleum ether/ethyl acetate = 10:1) as a liquid. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.45–7.21 (m, 11H; Ar-H and CH=C), 6.73 (s, 1H; NH), 4.53–4.35 (m, 2H; CH_2Ph), 3.79 (q, *J* = 7.2 Hz, 1H; COCH), 1.40 (d, *J* = 7.2 Hz, 3H; CH_3), 1.23 (s, 6H; B(pin)), 1.19 ppm (s, 6H; B(pin)); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 174.7, 145.0, 138.5, 136.5, 129.1, 128.4, 128.2, 127.72, 127.69, 127.2, 83.8, 43.6, 42.0, 24.8, 24.2, 17.8 ppm; IR (neat): $\tilde{\nu}$ = 3367, 2977, 2932, 1658, 1495, 1451, 1364, 1313, 1264, 1132 cm^{-1}; MS (70 eV, EI): *m*/*z* (%): 391 (20.42) $[M]^+$, 91 (100); HRMS: *m*/*z* calcd for $C_{24}H_{30}BNO_3$: 391.2319 $[M]^+$; found: 391.2318.

*Preparation of (Z)-N-benzyl-2-octanyl-4-p-methoxyphenyl-3-(4,4,5,5-tetra-methyl-1,3,2-dioxaborolan-2-yl)butyl-3-(Z)-enamide (Z-**2j**)*: The reaction of bis(pinacolato)diboron (153.5 mg, 0.60 mmol), P(*p*-MeOC$_6$H$_4$)$_3$ (10.4 mg, 0.03 mmol), CuCl (2.5 mg, 0.025 mmol), K_2CO_3 (13.5 mg, 0.10 mmol), **1j** (198.1 mg, 0.50 mmol), *i*PrOH (78 μL, *d* = 0.784 g mL^{-1}, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2j** (215.2 mg, 82%) after purification (petroleum ether/ethyl acetate = 10:1) as a white solid; m.p. 81–82 °C (*n*-hexane/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.50 (d, *J* = 8.7 Hz, 2H; Ar-H), 7.40 (s, 1H; Ar-H), 7.35–7.21 (m, 6H; Ar-H), 6.94–6.88 (m, 2H; CH=C and NH), 4.54–4.39 (m, 2H; CH_2Ph), 3.81 (s, 3H; OMe), 3.66 (dd, J_1 = 9.3 Hz, J_2 = 6.6 Hz, 1H; COCH), 1.98–1.68 (m, 2H; CH_2), 1.32–1.00 (m, 24H; 6×CH_2 and B(pin)), 0.85 ppm (t, *J* = 6.8 Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 174.9, 159.2, 146.3, 138.6, 131.1, 129.1, 128.5, 127.8, 127.2, 113.6, 83.8, 55.1, 48.8, 43.6, 32.6, 31.7, 29.3, 29.1, 27.7, 24.8, 24.1, 22.6, 14.0 ppm; IR (neat): $\tilde{\nu}$ = 3335, 2924, 2851, 1639, 1604, 1509, 1455, 1379, 1344, 1312, 1285, 1250, 1209, 1175, 1133, 1037 cm^{-1}; MS (70 eV, EI): *m*/*z* (%): 519 (3.34) $[M]^+$, 91 (100); elemental analysis calcd (%) for $C_{32}H_{46}BNO_4$: C 73.98, H 8.92, N 2.70; found: C 73.96, H 9.05, N 2.75.

*Preparation of (Z)-N-benzyl-2-butyl-4-thenyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)butyl-3-(Z)-enamide (Z-**2k**)*: The reaction of bis(pinacolato)diboron (153.0 mg, 0.60 mmol), P(*p*-MeOC$_6$H$_4$)$_3$ (10.8 mg, 0.03 mmol), CuCl (2.6 mg, 0.025 mmol), K_2CO_3 (13.7 mg, 0.10 mmol), **1k** (154.6 mg, 0.50 mmol), *i*PrOH (78 μL, *d* = 0.784 g mL^{-1}, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2k** (140.5 mg, 64%) after purification (petroleum ether/ethyl acetate = 15:1) as a liquid. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.48–7.43 (m, 2H; Ar-H), 7.36–7.18 (m, 6H; Ar-H), 7.04 (dd, J_1 = 5.0 Hz, J_2 = 3.8 Hz, 1H; CH=C), 6.92 (t, *J* = 5.3 Hz, 1H; NH), 4.48 (dd, J_1 = 14.7 Hz, J_2 = 5.4 Hz, 1H; CH*H*Ph), 4.37 (dd, J_1 = 14.7 Hz, J_2 = 5.1 Hz, 1H; C*H*HPh), 3.92 (dd, J_1 = 9.5 Hz, J_2 = 5.9 Hz, 1H; COCH), 2.12–1.97 (m, 1H; CH*H*), 1.96–1.80 (m, 1H; C*H*H), 1.37–1.08 (m, 16H; 2×CH_2 and B(pin)), 0.84 ppm (t, *J* = 6.8 Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 173.6, 139.2, 138.4, 137.5, 130.7, 128.3, 127.6, 127.29, 127.26, 127.0, 83.8, 49.1, 43.5, 31.7, 29.9, 24.7, 24.0, 22.4, 13.8 ppm; IR (neat): $\tilde{\nu}$ = 3374, 2930, 2860, 1666, 1588, 1508, 1369, 1341, 1312, 1216, 1141 cm^{-1}; MS (70 eV, EI): *m*/*z* (%): 439 (17.52) $[M]^+$, 91 (100); HRMS: *m*/*z* calcd for $C_{25}H_{34}BNO_3S$: 439.2352 $[M]^+$; found: 439.2354.

*Preparation of (Z)-N-benzyl-2-tosyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pent-3-(Z)-enamide (Z-**2l**)*: The reaction of bis(pinacolato)diboron (152.4 mg, 0.60 mmol), P(*p*-MeOC$_6$H$_4$)$_3$ (10.6 mg, 0.03 mmol), CuCl (2.4 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1l** (138.6 mg, 0.50 mmol), *i*PrOH (78 μL, *d* = 0.784 g mL^{-1}, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2l** (134.7 mg, 67%) after purification (petroleum ether/ethyl acetate = 10:1) as a liquid. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.33–7.18 (m, 7H; Ar-H), 7.21 (d, *J* = 8.1 Hz, 2H; Ar-H), 6.68–6.58 (m, 2H; CH=C and NH), 4.74 (s, 1H; COCH), 4.55 (dd, J_1 = 15.2 Hz, J_2 = 6.2 Hz, 1H; C*H*HPh), 4.37 (dd, J_1 = 14.9 Hz, J_2 = 5.0 Hz, 1H; CH*H*Ph), 2.29 (s, 3H; PhCH_3), 1.80 (d, *J* = 6.6 Hz, 3H; =CCH_3), 1.10 ppm (s, 12H; B(pin)); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 173.1, 143.2, 138.5, 136.8, 136.1, 129.1, 128.3, 128.2, 127.5, 127.0, 83.5, 52.6, 43.6, 24.6, 24.2, 20.9, 14.6 ppm; IR (neat): $\tilde{\nu}$ = 3350, 2977, 1656, 1513, 1381, 1335, 1307, 1146, 1133 cm^{-1}; MS (70 eV, EI): *m*/*z* (%): 405 (15.41) $[M]^+$, 91 (100); HRMS: *m*/*z* calcd for $C_{25}H_{32}BNO_3$: 405.2475 $[M]^+$; found: 405.2477.

*Preparation of (Z)-N-benzyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pent-3-(Z)-enamide (Z-**2m**)*: The reaction of bis(pinacolato)diboron (152.6 mg, 0.60 mmol), P(*p*-MeOC$_6$H$_4$)$_3$ (10.6 mg, 0.03 mmol), CuCl (2.4 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1m** (94.0 mg, 0.50 mmol), *i*PrOH (78 μL, *d* = 0.784 g mL^{-1}, 60.1 mg, 1.00 mmol), and THF (2 mL) afforded *Z*-**2m** (142.3 mg, 90%) after purification (petroleum ether/ethyl acetate = 5:1) as a liquid. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.35–7.21 (m, 5H; Ar-H), 6.65 (q, *J* = 6.9 Hz, 1H; =CH), 6.37 (br s, 1H; NH), 4.40 (d, *J* = 5.4 Hz, 2H; CH_2Ph), 3.16 (s, 2H; $COCH_2$), 1.84 (d, *J* = 6.6 Hz, 3H; =CCH_3), 1.16 ppm (s, 12H; B(pin)); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 171.2, 145.5, 138.5, 128.5, 127.7, 127.2, 83.7, 43.5, 36.4, 24.6, 14.8 ppm; IR (neat): $\tilde{\nu}$ = 3378, 2978, 1654, 1533, 1386, 1370, 1336, 1306, 1143, 1110, 1030 cm^{-1}; MS (70 eV, EI): *m*/*z* (%): 315 (20.21) $[M]^+$, 91 (100); HRMS: *m*/*z* calcd for $C_{18}H_{26}BNO_3$: 315.2006 $[M]^+$; found: 315.2008.

*Preparation of N-benzyl-4-methyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pent-3-enamide (**2n**)*: The reaction of bis(pinacolato)diboron (154.0 mg, 0.60 mmol), P(*p*-MeOC$_6$H$_4$)$_3$ (10.7 mg, 0.03 mmol), CuCl (2.5 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1n** (100.7 mg, 0.50 mmol), *i*PrOH (78 μL, *d* = 0.784 g mL^{-1}, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded **2n** (159.7 mg, 97%) after purification (petroleum ether/ethyl acetate = 7:1) as a white solid; m.p. 77–78 °C (petroleum ether/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.34–7.20 (m, 5H; Ar-H), 6.44 (br s, 1H; NH), 4.40 (d, *J* = 5.7 Hz, 2H; CH_2Ph), 3.13 (s, 2H; $COCH_2$), 2.05 (s, 3H; =CCH_3), 1.86 (s, 3H; =CCH_3), 1.18 ppm (s, 12H; B(pin)); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 172.1, 154.6, 138.6, 128.4, 127.5, 127.1, 83.2, 43.3, 38.9, 24.62, 24.59, 22.0 ppm; IR (neat): $\tilde{\nu}$ = 3232, 2978, 2929, 2858, 1640, 1522, 1371, 1351, 1143, 1070 cm^{-1}; MS (70 eV, EI): *m*/*z* (%): 329 (19.62) $[M]^+$, 91 (100); elemental analysis calcd (%) for $C_{19}H_{28}BNO_3$: C 69.31, H 8.57, N 4.25; found: C 69.44, H 8.45, N 4.17.

*Preparation of N-phenyl-4-methyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pent-3-enamide (**2o**)*: The reaction of bis(pinacolato)diboron (153.0 mg, 0.60 mmol), P(*p*-MeOC$_6$H$_4$)$_3$ (10.8 mg, 0.03 mmol), CuCl (2.4 mg, 0.025 mmol), K_2CO_3 (13.8 mg, 0.10 mmol), **1o** (94.0 mg, 0.50 mmol), *i*PrOH (78 μL, *d* = 0.784 g mL^{-1}, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded **2o** (157.7 mg, 100%) after purification (petroleum ether/ethyl acetate = 7:1) as a white solid; m.p. 99–100 °C (*n*-hexane/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.25 (br s, 1H; NH), 7.50 (d, *J* = 7.8 Hz, 2H; Ar-H), 7.33–7.24 (m, 2H; Ar-H), 7.08–7.01 (m, 1H; Ar-H), 3.20 (s, 2H; $COCH_2$), 2.07 (s, 3H; =CCH_3), 1.93 (s, 3H; =CCH_3), 1.32 ppm (s, 12H; B(pin)); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 170.5, 155.6, 138.7, 128.8, 123.4, 119.1, 83.5, 40.3, 24.8, 24.6, 22.1 ppm; IR (neat): $\tilde{\nu}$ = 3293, 2930, 2856, 1734, 1661, 1638, 1599, 1532, 1441, 1370, 1296, 1252, 1145, 1031, 1004 cm^{-1}; MS (70 eV, EI): *m*/*z* (%): 315 (85.49) $[M]^+$, 77

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(100); elemental analysis calcd (%) for $C_{18}H_{26}BNO_3$: C 68.59, H 8.31, N 4.44; found: C 68.74, H 8.25, N 4.33.

*Preparation of N-butyl-4-methyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pent-3-enamide (**2p**)*: The reaction of bis(pinacolato)diboron (152.4 mg, 0.60 mmol), P(*p*-$MeOC_6H_4)_3$ (10.6 mg, 0.03 mmol), CuCl (2.6 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1p** (83.8 mg, 0.50 mmol), *i*PrOH (78 μL, $d=0.784\ g\,mL^{-1}$, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded **2p** (144.9 mg, 98%) after purification (petroleum ether/ethyl acetate = 7:1) as a liquid. ^1H NMR (300 MHz, $CDCl_3$): $\delta=$ 6.01 (br s, 1H; NH), 3.24–3.15 (m, 2H; CH_2), 3.06 (s, 2H; $COCH_2$), 2.05 (s, 3H; $=CCH_3$), 1.83 (s, 3H; $=CCH_3$), 1.50–1.23 (m, 16H; $2\times CH_2$ and B-(pin)), 0.91 ppm (t, $J=7.2$ Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=$172.0, 153.9, 83.1, 38.8, 31.5, 24.6, 24.5, 21.7, 19.8, 13.6 ppm; IR (neat): $\tilde{\nu}=$3307, 2976, 2930, 2873, 1647, 1541, 1372, 1354, 1294, 1221, 1145, 1063 cm^{-1}; MS (70 eV, EI): *m/z* (%): 295 (35.12) $[M]^+$, 139 (100); HRMS: *m/z* calcd for $C_{16}H_{30}BNO_3$: 295.2319 $[M]^+$; found: 295.2320.

*Preparation of (Z)-N-benzyl-4-methyl-5,5-dimethyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)hex-3-(Z)-enamide (Z-**2q**)*: The reaction of bis-(pinacolato)diboron (152.4 mg, 0.60 mmol), P(*p*-$MeOC_6H_4)_3$ (11.0 mg, 0.03 mmol), CuCl (2.5 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1q** (121.7 mg, 0.50 mmol), *i*PrOH (78 μL, $d=0.784\ g\,mL^{-1}$, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2q** (130.2 mg, 70%) after purification (petroleum ether/ethyl acetate = 7:1) as a pale-yellow solid; m.p. 108–109 °C (*n*-hexane/ethyl ether); ^1H NMR (300 MHz, $CDCl_3$): $\delta=$7.34–7.21 (m, 5H; Ar-H), 6.35 (br s, 1H; NH), 4.44 (d, $J=5.4$ Hz, 2H; CH_2Ph), 3.36 (s, 2H; $COCH_2$), 1.93 (s, 3H; $=CCH_3$), 1.21 (s, 12H; B(pin)), 1.15 ppm (s, 9H; $=(CH_3)_3$); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=$ 172.0, 158.9, 138.3, 128.3, 127.6, 127.0, 83.2, 43.3, 39.4, 37.4, 30.2, 24.5, 22.4 ppm; IR (neat): $\tilde{\nu}=$3264, 2974, 2906, 1648, 1546, 1475, 1455, 1366, 1348, 1288, 1231, 1142, 1063, 1040, 1008 cm^{-1}; MS (70 eV, EI): *m/z* (%): 371 (11.26) $[M]^+$, 91 (100); elemental analysis calcd (%) for $C_{22}H_{34}BNO_3$: C 71.16, H 9.23, N 3.77; found: C 71.13, H 9.28, N 3.78.

*Preparation of (Z)-N-benzyl-4-phenyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pent-3-(Z)-enamide (Z-**2r**)*: The reaction of bis(pinacolato)-diboron (153.0 mg, 0.60 mmol), P(*p*-$MeOC_6H_4)_3$ (10.5 mg, 0.03 mmol), CuCl (2.6 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1r** (132.0 mg, 0.50 mmol), *i*PrOH (78 μL, $d=0.784\ g\,mL^{-1}$, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded *Z*-**2r** (148.6 mg, 76%) after purification (petroleum ether/ethyl acetate = 7:1) as a white solid; m.p. 105–106 °C (*n*-hexane/ethyl ether); ^1H NMR (300 MHz, $CDCl_3$): $\delta=$7.34–7.18 (m, 8H; Ar-H), 7.10 (d, $J=7.5$ Hz, 2H; Ar-H), 6.08 (br s, 1H; NH), 4.37 (d, $J=6.0$ Hz, 2H; CH_2), 2.97 (s, 2H; $COCH_2$), 2.29 (s, 3H; $=CCH_3$), 1.25 ppm (s, 12H; B(pin)); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=$171.9, 154.7, 143.5, 138.4, 128.3, 128.1, 127.6, 127.1, 126.9, 126.7, 83.3, 43.3, 39.9, 24.9, 24.6 ppm; IR (neat): $\tilde{\nu}=$3309, 2983, 2932, 1640, 1526, 1445, 1398, 1362, 1340, 1306, 1205, 1145, 1119, 1040 cm^{-1}; MS (70 eV, EI): *m/z* (%): 391 (15.35) $[M]^+$, 91 (100); elemental analysis calcd (%) for $C_{24}H_{30}BNO_3$: C 73.66, H 7.73, N 3.58; found: C 73.40, H 7.73, N 3.50.

*Preparation of N-benzyl-2,4-dimethyl-5,5-dimethyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pent-3-enamide (**2s**)*: The reaction of bis(pinacolato)diboron (152.4 mg, 0.60 mmol), P(*p*-$MeOC_6H_4)_3$ (10.4 mg, 0.03 mmol), CuCl (2.6 mg, 0.025 mmol), K_2CO_3 (13.9 mg, 0.10 mmol), **1s** (108.0 mg, 0.50 mmol), *i*PrOH (78 μL, $d=0.784\ g\,mL^{-1}$, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded **2s** (150.5 mg, 87%) after purification (petroleum ether/ethyl acetate = 10:1) as a white solid; m.p. 112–113 °C (*n*-hexane/ethyl ether); ^1H NMR (300 MHz, $CDCl_3$): $\delta=$7.38–7.18 (m, 5H; Ar-H), 6.54 (br s, 1H; NH), 4.48 (dd, $J_1=14.7$ Hz, $J_2=5.7$ Hz, 1H; C*H*H), 4.35 (dd, $J_1=15.0$ Hz, $J_2=5.1$ Hz, 1H; CH*H*), 3.44 (q, $J=$ 7.2 Hz, 1H; COCH), 1.93 (s, 3H; $=CCH_3$), 1.76 (s, 3H; $=CCH_3$), 1.46–1.10 ppm (m, 15H; CH_3 and B(pin)); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=$ 175.3, 147.5, 138.6, 128.2, 127.4, 126.9, 83.3, 43.3, 43.2, 25.0, 24.8, 24.5, 20.8, 17.3 ppm; IR (neat): $\tilde{\nu}=$3317, 2976, 2926, 2870, 1655, 1542, 1456, 1388, 1370, 1333, 1268, 1228, 1214, 1142, 1094 cm^{-1}; MS (70 eV, EI): *m/z* (%): 343 (27.23) $[M]^+$, 91 (100); elemental analysis calcd (%) for $C_{20}H_{30}BNO_3$: C 69.98, H 8.81, N 4.08; found: C 70.24, H 8.86, N 3.92.

Synthetic applications of borylated product Z-2d

*Preparation of (E)-N-benzyl-2-butyl-3-chloride-5-methyl-hex-3-(E)-enamide (**3**)*: To an oven-dried Schlenk tube equipped with a magnetic stirring bar were added sequentially $CuCl_2$ (81.0 mg, 0.60 mmol), *Z*-**2d** (80.4 mg, 0.20 mmol), MeOH (1 mL), and H_2O (1 mL). The mixture was stirred at 80 °C for 2.5 h and monitored by TLC. Upon completion, Et_2O (5 mL) was added to the resulting mixture, then the aqueous phase was extracted with Et_2O (3×5 mL). The combined organic layer was dried over anhydrous Na_2SO_4. Filtration, evaporation, and chromatography on silica gel (petroleum ether/ethyl acetate = 7:1) afforded **3** (55.6 mg, 90%) as a liquid. ^1H NMR (300 MHz, $CDCl_3$): $\delta=$7.37–7.22 (m, 5H; Ar-H), 6.14 (br s, 1H; NH), 5.72 (d, $J=10.2$ Hz, 1H; CH=C), 4.53–4.39 (m, 2H; CH_2Ph), 3.58 (dd, $J_1=10.2$ Hz, $J_2=4.8$ Hz, 1H; COCH), 2.68–2.50 (m, 1H; CH), 2.08–1.93 (m, 1H; CH*H*), 1.88–1.71 (m, 1H; C*H*H), 1.43–1.10 (m, 4H; $2\times CH_2$), 1.00 (d, $J=6.6$ Hz, 3H; CH_3), 0.99 (d, $J=6.6$ Hz, 3H; CH_3), 0.91 ppm (t, $J=7.1$ Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=$170.4, 140.4, 138.1, 129.5, 128.6, 127.4, 127.3, 49.3, 43.6, 29.2, 28.6, 28.5, 22.8, 22.7, 22.4, 13.9 ppm; IR (neat): $\tilde{\nu}=$3304, 2959, 2928, 2870, 1651, 1542, 1455, 1362, 1220, 1173, 1029 cm^{-1}; MS (70 eV, EI): *m/z* (%): 309 (0.87) $[M(^{37}Cl)]^+$, 307 (2.73) $[M(^{35}Cl)]^+$, 91 (100); HRMS: *m/z* calcd for $C_{18}H_{26}{}^{35}ClNO$: 307.1703 $[M]^+$; found: 307.1701.

*Preparation of (E)-N-benzyl-2-butyl-3-boromide-5-methyl-hex-3-(E)-enamide (**4**)*: The reaction of $CuBr_2$ (135.0 mg, 0.60 mmol), *Z*-**2d** (79.0 mg, 0.20 mmol), MeOH (1 mL), and H_2O (1 mL) afforded **4** (56.3 mg, 81%) after purification (petroleum ether/ethyl acetate = 7:1) as a liquid. ^1H NMR (300 MHz, $CDCl_3$): $\delta=$7.37–7.23 (m, 5H; Ar-H), 6.11 (br s, 1H; NH), 6.00 (d, $J=9.9$ Hz, 1H; CH=C), 4.53–4.39 (m, 2H; CH_2Ph), 3.52 (dd, $J_1=10.2$ Hz, $J_2=4.5$ Hz, 1H; COCH), 2.72–2.53 (m, 1H; CH), 2.09–1.93 (m, 1H; CH*H*), 1.82–1.68 (m, 1H; C*H*H), 1.43–1.09 (m, 4H; $2\times$ CH_2), 1.00 (d, $J=6.6$ Hz, 3H; CH_3), 0.98 (d, $J=6.9$ Hz, 3H; CH_3), 0.91 ppm (t, $J=7.2$ Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=$ 170.4, 144.8, 138.1, 128.6, 127.45, 127.36, 122.0, 50.0, 43.7, 29.8, 29.5, 29.2, 22.6, 22.5, 22.4, 13.9 ppm; IR (neat): $\tilde{\nu}=$3306, 2958, 2928, 2869, 1651, 1538, 1455, 1362, 1218, 1168, 1153, 1029 cm^{-1}; MS (70 eV, EI): *m/z* (%): 353 (1.24) $[M(^{81}Br)]^+$, 351 (1.12) $[M(^{79}Br)]^+$, 91 (100); HRMS: *m/z* calcd for $C_{18}H_{26}{}^{79}BrNO$: 351.1198 $[M]^+$; found: 351.1196.

*Preparation of N-benzyl-2-butyl-5-methyl-3-oxohexanamide (**5**)*: To an oven-dried Schlenk tube equipped with a magnetic stirring bar were added *Z*-**2d** (79.9 mg, 0.20 mmol) and THF (1 mL), then NaOH (1 mL, C = 3 M) and H_2O_2 (1 mL, 30% aq.) were added dropwise at 0 °C within 10 and 5 min, respectively, and the resulting mixture was stirred at RT. Upon completion, the reaction was quenched with saturated $Na_2S_2O_3$ (1 mL) and NaOH (1 mL, C = 1 M), extracted with ethyl acetate (3× 10 mL), washed with brine, and dried over Na_2SO_4. Filtration, evaporation, and chromatography on silica gel (petroleum ether/ethyl acetate = 7:1) afforded **5** (47.1 mg, 81%) as a white solid; m.p. 96–97 °C (*n*-hexane/ethyl ether); ^1H NMR (300 MHz, $CDCl_3$): $\delta=$7.36–7.18 (m, 5H; Ar-H), 6.76 (br s, 1H; NH), 4.51–4.33 (m, 2H; CH_2Ph), 3.40 (t, $J=7.4$ Hz, 1H; COCHCO), 2.51–2.32 (m, 2H; CH_2), 2.13 (hept, $J=6.6$ Hz, 1H; CH), 1.89–1.72 (m, 2H; CH_2), 1.40–1.18 (m, 4H; $2\times CH_2$), 0.93–0.82 ppm (m, 9H; $3\times CH_3$); ^{13}C NMR (75.4 MHz, $CDCl_3$): $\delta=$209.4, 168.5, 138.0, 128.6, 127.5, 127.4, 61.0, 51.5, 43.5, 30.6, 29.5, 23.8, 22.4, 22.3, 13.7 ppm; IR (neat): $\tilde{\nu}=$3291, 2956, 2928, 2862, 1707, 1667, 1641, 1558, 1523, 1496, 1456, 1405, 1365, 1344, 1293, 1244, 1219, 1145, 1082, 1042, 1030, 1016 cm^{-1}; MS (70 eV, EI): *m/z* (%): 289 (12.87) $[M]^+$, 91 (100); elemental analysis calcd (%) for $C_{18}H_{27}NO_2$: C 74.70, H 9.40, N 4.84; found: C 74.37, H 9.53, N 4.75.

*Preparation of (Z)-N-benzyl-2-butyl-3-phenyl-5-methyl-hex-3-(Z)-enamide (**6**)*: To an oven-dried Schlenk tube equipped with a magnetic stirring bar were added sequentially $[Pd(PPh_3)_4]$ (11.7 mg, 0.01 mmol), K_3PO_4 (127.4 mg, 0.6 mmol), *Z*-**2d** (79.5 mg, 0.20 mmol), toluene (1 mL), and PhI (81.8 mg, 0.4 mmol) under argon. The resulting mixture was stirred at 85 °C for 24 h and monitored by TLC. Upon completion, the resulting mixture was filtered through a short column of silica gel eluting with Et_2O (3×20 mL) and concentrated. The residue was purified by chromatography on silica gel (petroleum ether/ethyl acetate = 10:1) to afford **6** (67.2 mg, 96%) as a liquid. ^1H NMR (300 MHz, $CDCl_3$): $\delta=$7.33–7.17 (m, 10H; Ar-H), 6.23 (t, $J=5.1$ Hz, 1H; NH), 5.69 (d, $J=9.9$ Hz, 1H; = CH), 4.51 (dd, $J_1=14.6$ Hz, $J_2=5.9$ Hz, 1H; C*H*HPh), 4.40 (dd, $J_1=$ 14.4 Hz, $J_2=5.4$ Hz, 1H; CH*H*Ph), 3.61 (dd, $J_1=10.1$ Hz, $J_2=4.7$ Hz, 1H; COCH), 2.81–2.63 (m, 1H; CH), 2.22–2.07 (m, 1H; CH*H*), 1.56–1.40 (m,

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1H; C*H*H), 1.32–1.07 (m, 4H; 2×CH_2), 1.02 (d, J=6.3 Hz, 3H; CH_3), 0.99 (d, J=6.6 Hz, 3H; CH_3), 0.78 ppm (t, J=6.9 Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ=172.8, 142.0, 140.8, 138.2, 136.5, 128.6, 128.1, 127.8, 127.4, 127.2, 127.0, 47.2, 43.9, 29.9, 28.2, 27.9, 23.0, 22.9, 22.4, 13.9 cm^{-1}; IR (neat): $\tilde{\nu}$=3337, 2956, 2927, 2867, 1654, 1508, 1455, 1361, 1252, 1029 cm^{-1}; MS (70 eV, EI): m/z (%): 349 (4.84) $[M]^+$, 292 (100); HRMS: m/z calcd for $C_{24}H_{31}NO$: 349.2406 $[M]^+$; found: 349.2407.

Mechanistic study

*Preparation of N-benzyl-2-deutero-4-methyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pent-3-enamide ([D]-**2n**)*: The reaction of bis(pinacolato)diboron (152.6 mg, 0.60 mmol), P(*p*-$MeOC_6H_4$)$_3$ (10.7 mg, 0.03 mmol), CuCl (2.5 mg, 0.025 mmol), K_2CO_3 (13.8 mg, 0.10 mmol), **1n** (100.6 mg, 0.50 mmol), *i*PrOD (69.0 mg, 1.00 mmol), and Et_2O (2 mL) afforded **2n**-*d* (155.8 mg, 94%, D%=50%) after purification (petroleum ether/ethyl acetate=7:1) as a white solid; m.p. 78–79°C (petroleum ether/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$): δ=7.35–7.21 (m, 5H; Ar-H), 6.42 (br s, 1H; NH), 4.41 (d, J=5.7 Hz, 2H; CH_2), 3.13 (d, J=5.4 Hz, 1H; COCHD), 2.05 (s, 3H; =CCH_3), 1.86 (s, 3H; =CCH_3), 1.18 ppm (s, 12H; B(pin)); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ=172.1, 154.65, 138.6, 128.4, 127.5, 127.1, 83.2, 43.3, 38.6 (t, J=20.3 Hz), 24.6, 22.0; peaks at δ=154.70 and 38.9 ppm arise from undeuterated product; IR (neat): $\tilde{\nu}$=3257, 3085, 2976, 2927, 1637, 1561, 1496, 1454, 1370, 1352, 1333, 1295, 1273, 1220, 1146, 1110, 1070, 1046, 1026 cm^{-1}; MS (70 eV, EI): m/z (%): 330 (10.37) $[M(^2D)]^+$, 329 (10.28) $[M(^1H)]^+$, 91 (100); elemental analysis calcd (%) for $C_{19}H_{27}DBNO_3$: C 69.10, H 8.85, N 4.24; found: C 69.32, H 8.61, N 4.17.

*Preparation of N-diethyl-4-methyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pent-3-enamide (**2t**)*: The reaction of bis(pinacolato)diboron (153.0 mg, 0.60 mmol), P(*p*-$MeOC_6H_4$)$_3$ (10.6 mg, 0.03 mmol), CuCl (2.5 mg, 0.025 mmol), K_2CO_3 (13.8 mg, 0.10 mmol), **1t** (83.5 mg, 0.50 mmol), *i*PrOH (78 μL, d=0.784 g mL^{-1}, 60.1 mg, 1.00 mmol), and Et_2O (2 mL) afforded **2t** (105.5 mg, 72%) after purification (petroleum ether/ethyl acetate=3:1) as a white solid; m.p. 46–47°C (*n*-hexane/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$): δ=3.43–3.28 (m, 4H; 2×CH_2), 3.20 (s, 2H; $COCH_2$), 1.98 (s, 3H; =CCH_3), 1.70 (s, 3H; =CCH_3), 1.27 (s, 12H; B(pin)), 1.19 (t, J=7.1 Hz, 3H; CH_3), 1.12 ppm (t, J=7.1 Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ=172.9, 143.9, 81.9, 42.1, 40.6, 35.9, 25.3, 23.9, 21.7, 14.0, 13.0; IR (neat): $\tilde{\nu}$=2974, 2930, 2873, 1724, 1627, 1460, 1445, 1430, 1372, 1353, 1319, 1286, 1272, 1255, 1220, 1147, 1136, 1066 cm^{-1}; MS (70 eV, EI): m/z (%): 295 (41.01) $[M]^+$, 100 (100); elemental analysis calcd (%) for $C_{16}H_{30}BNO_3$: C 65.09, H 10.24, N 4.74; found: C 64.93, H 10.34, N 4.68.

*Preparation of N-diethyl-2-deuter-4-methyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pent-3-enamide ([D]-**2t**)*: The reaction of bis(pinacolato)diboron (150.0 mg, 0.60 mmol), P(*p*-$MeOC_6H_4$)$_3$ (10.5 mg, 0.03 mmol), CuCl (2.5 mg, 0.025 mmol), K_2CO_3 (13.7 mg, 0.10 mmol), **1t** (81.4 mg, 0.50 mmol), *i*PrOD (69.0 mg, 1.00 mmol), and Et_2O (2 mL) afforded **2t**-*d* (102.0 mg, 71%, D%=84%) after purification (petroleum ether/ethyl acetate=3:1) as a white solid; m.p. 47–48°C (*n*-hexane/ethyl ether); ^{1}H NMR (300 MHz, $CDCl_3$): δ=3.43–3.28 (m, 4H; 2×CH_2), 3.18 (d, J=5.7 Hz, 1H; COCHD), 1.98 (s, 3H; =CCH_3), 1.71 (s, 3H; =CCH_3), 1.27 (s, 12H; B(pin)), 1.19 (t, J=7.1 Hz, 3H; CH_3), 1.12 ppm (t, J=7.1 Hz, 3H; CH_3); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ=172.78, 144.08, 81.81, 42.0, 40.5, 35.4 (t, J=19.1 Hz), 25.2, 23.9, 21.6, 13.9, 12.9 ppm; the peaks at δ=172.84, 144.03, 81.77, 35.8 ppm arise from undeuterated product; IR (neat): $\tilde{\nu}$=2976, 2930, 1626, 1477, 1460, 1443, 1429, 1371, 1352, 1286, 1218, 1147, 1072 cm^{-1}; MS (70 eV, EI): m/z (%): 296 (53.20) $[M(^2D)]^+$, 295 (49.34) $[M(^1H)]^+$, 72 (100); elemental analysis calcd (%) for $C_{16}H_{29}DBNO_3$: C 64.87, H 10.55, N 4.73; found: C 65.10, H 10.39, N 4.67.

Acknowledgements

Financial support from the National Basic Research Program of China (No. 2011CB808700) and the National Natural Science Foundation of China (No. 21232006 and 21102164) is greatly appreciated. We thank Mr Rong Zeng in this group for reproducing the results of Table 2 (entries 3 and 13) and Scheme 2 (*Z*-**2r**) presented in this study.

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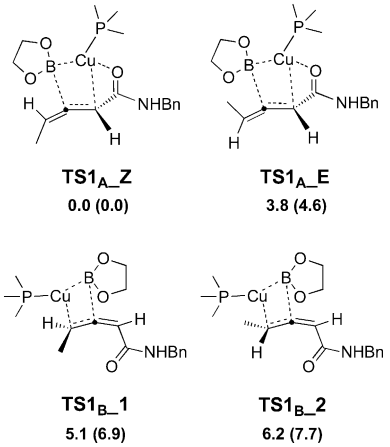
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[21] The other two transition structures associated with **TS1$_B$**, which are denoted as **TS1$_B$_1** and **TS1$_B$_2**, are both higher in free energy (and electronic energy) than **TS1$_A$_*Z*** and **TS1$_A$_*E***. The relative free energies and electronic energies (in parentheses) of the four transition structures are shown in kcal mol^{-1}. For the details see the Supporting Information.



Received: August 8, 2012
Revised: December 12, 2012
Published online: April 4, 2013

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